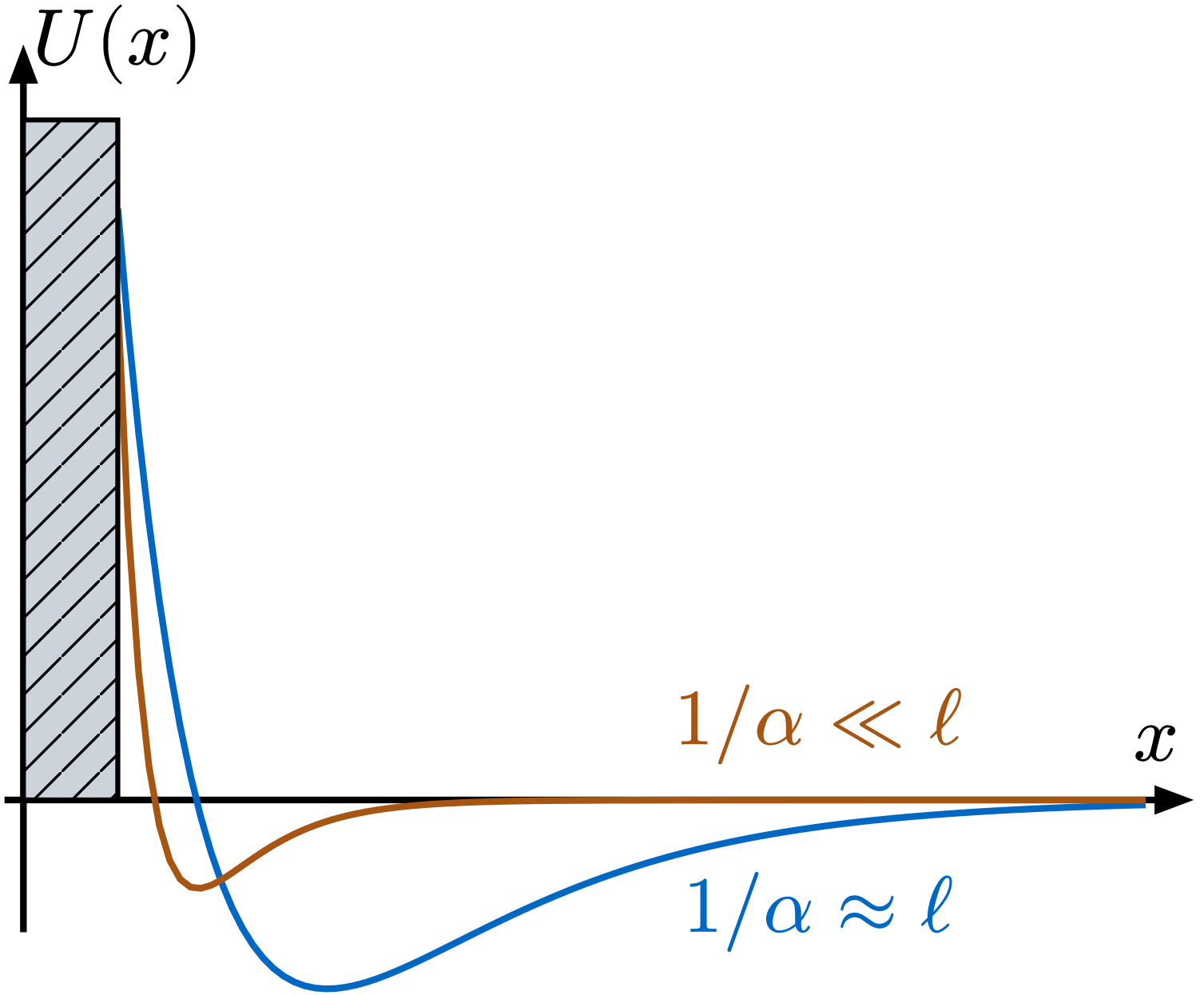
$U(x)$
x
$1/\alpha \ll \ell$
$1/\alpha \approx \ell$